Operating-Regime-Aware Anomaly Detection for Gas-Turbine Time Series

1. Problem statement

Anomaly detection in gas-turbine systems is difficult because sensor behaviour changes substantially across operating conditions such as load, speed, and control state. A pattern that is normal at high load may appear abnormal at low load, and vice versa. This causes global anomaly-detection models with a single alarm rule to produce unstable behaviour, including false alarms during normal regime changes and missed detections during genuine degradation.

We will develop a rigorous anomaly-detection framework for multivariate gas-turbine time series that explicitly accounts for operating regime. Our aim is not to propose a radically new deep-learning architecture, but to show that anomaly detection becomes more reliable when normal behaviour is learned relative to operating condition and alarm thresholds are calibrated accordingly. The focus will therefore be on robustness, early warning, and false-alarm control under nonstationary operating conditions.

So, the central hypothesis is that regime-aware normality modelling and regime-specific score calibration will improve the reliability of anomaly detection compared with global models that ignore operating context.

2. Data

Our primary dataset should be the Naval dataset, which seems to be well suited to this study because it represents a propulsion system containing a gas turbine and includes multivariate measurements together with degradation-related variables. This provides an application-specific test bed for anomaly detection under changing operating conditions.

Our secondary option is the C-MAPSS dataset. Its multi-condition subsets will be useful to test whether the proposed framework generalises beyond a single system and remains effective when operating regimes vary.

3. Methodology

We should aim to model normal behaviour and detect anomalies through deviations from expected sensor dynamics. A lightweight sequence predictor, such as a TCN, should work. We can train on healthy data only to predict the next sensor state from recent history. Anomaly detection is then based on the discrepancy between predicted and observed behaviour.

Our contribution does not lie in the predictor alone, but in the treatment of operating regime during modelling and calibration. Operating regimes are defined from known operating variables or from a clustering procedure fitted on the training set only. Rather than relying on a single global notion of normality, the predictor is conditioned on operating regime, allowing expected behaviour to vary with load and operating state.

Anomaly scores are then calibrated separately within each regime. Since residual distributions differ across operating conditions, raw residuals are converted into regime-normalised anomaly scores, for example using covariance-normalised residual distances. Regime-specific calibration is then used to determine alarm thresholds on validation data, improving score consistency and reducing false alarms caused by normal regime changes.

We will evaluate the performance against baselines, including global forecasting models, reconstruction-based autoencoders, and classical unsupervised detectors.

4. Core contribution

We should position our work as a rigorous applied anomaly-detection study, not as a claim of a wholly new deep architecture. Its contribution is to show that, in gas-turbine monitoring, explicitly accounting for operating regime materially improves the stability and usefulness of anomaly alarms.

5. Novelty

I believe we have the following novelties (but please check the literature):

1. **Operating-regime-aware normality modelling**
   Normal behaviour is learned relative to operating condition rather than globally, which directly addresses non-stationarity in gas-turbine sensor data.
2. **Regime-specific anomaly score calibration**
   Alarm thresholds are adapted to the residual characteristics of each operating regime, reducing false alarms caused by normal condition changes.
3. **Rigorous evaluation focused on reliability**
   The framework is assessed using leakage-safe splits, strong baselines, event-level metrics, false alarm analysis, and detection delay, which makes the study methodologically stronger than standard benchmark-only comparisons.